divide.

3. **Environmental Impact**:
    - **Challenge**: AI systems, particularly large language models, consume significant computational resources and energy.
    - **Solutions**:
        - Optimize algorithms to reduce energy consumption.
        - Explore green AI initiatives that prioritize sustainability, such as using renewable energy for data centers.
    - **Example**: Implementing pruning and quantization techniques to reduce the size of neural networks without sacrificing performance.
4. **Legal and Regulatory Compliance**:
    - **Necessity**: Stay updated on regional and international AI regulations, including GDPR, CCPA, and emerging frameworks like the EU's AI Act.
    - **Action Steps**:
        - Document compliance efforts and maintain audit trails.
        - Conduct regular reviews to ensure adherence to legal standards.
    - **Example**: Ensuring transparency by providing users with detailed documentation on how their data is used and processed.
5. **Continuous Monitoring**:
    - **Rationale**: AI systems operate in dynamic environments and can be influenced by changing societal norms, values, or external threats.
    - **Approach**: